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Operating System Group Assignment

Group -11_Sec-A

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Why we chose these four

When you open a laptop or pick up a phone, you're not just holding a device. You're holding the work of dozens of companies. Each one specializes in something different. For this assignment, we picked one company from four important hardware categories. We wanted companies that are influential, interesting, and maybe a bit underappreciated. Here they are:

- **Qualcomm** – designs the brain inside most Android phones and many new Windows laptops.
- **Acer** – makes the whole computer, from the keyboard to the cooling fan.
- **G.Skill** – turns standard memory chips into ultra-fast RAM for gamers and creators.
- **ADATA** – quietly builds the SSDs that make your computer feel snappy.

Let's get to know them, not as logos on a spec sheet, but as real teams of engineers who actually care about what they build.

Task 1: Qualcomm – The wireless chip pioneer

Figure 1: Qualcomm Corporate Logo



Who they are and what they do

Qualcomm is based in sunny San Diego, California. It was founded in 1985. Today, Cristiano Amon is the CEO. The company's revenue was about \$45 billion in 2025 – that's a lot of chips. But here's the thing: Qualcomm doesn't actually make the chips themselves. They design them and then send the blueprints to factories like TSMC. This is called a "fabless" model. It means they focus entirely on invention, not on running factories.

Qualcomm's big idea was wireless. They made CDMA work, which became the heart of 3G networks. Then they moved to OFDMA for 4G and 5G. Along the way, they also got very good at building tiny, powerful processors.

What they make

The most famous product is the **Snapdragon** processor. You've probably heard of it – it's the engine inside most flagship Android phones.

Figure 2: Qualcomm Snapdragon 782G System-on-Chip



Snapdragon 8 series (for phones)

These are the chips inside flagship Android phones. The latest, Snapdragon 8 Gen 3, has eight cores: one very fast core, five fast cores, and two slow but very efficient cores. Why? Because your phone doesn't always need full power. When you're just checking email, it uses the small cores and saves battery. The AI performance is up to 45 TOPS – that's trillions of operations per second, which lets your phone do things like remove background noise in real time. Pretty cool, right?

Snapdragon X series (for laptops)

Qualcomm wants to beat Apple and Intel in laptops. The Snapdragon X Elite has 12 cores (8 fast, 4 efficient). It runs Windows on ARM without emulation. The battery life is excellent, and the built-in AI engine can handle tasks like real-time translation. We've seen reviews where the fan barely turns on during a whole day of work. That's the kind of experience Qualcomm is aiming for – no noise, no heat, just work.

Automotive chips

They also make chips for cars. The Snapdragon Ride Flex handles both self-driving systems and the entertainment screen. BMW, Mercedes, and Volkswagen all use Qualcomm inside their newer models. So your next car might have a Snapdragon brain too.

How this matters to an operating system

Think of the OS as a traffic cop. It has to decide which core gets which task. Qualcomm provides special drivers that tell the OS: "Hey, this core is for heavy lifting, this one is for background noise." Without those drivers, the battery would drain fast, and the phone would get hot. On Windows on ARM, the OS works closely with Qualcomm's scheduler to keep everything smooth. It's a beautiful partnership when it works well.

A quick comparison with competitors

- **Apple** – faster per watt, but only works in Apple devices. Sorry, no Android love.
- **MediaTek** – cheaper and decent, but weaker AI.

- **Intel** – great x86 software support, but uses more power.
- **AMD** – strong graphics, but no built-in 5G.

Qualcomm's strength is combining a good CPU, a good GPU, and a world-class modem all on one chip. That's hard to beat.

Task 2: Acer – The computer builder for everyone

Figure 3: Acer Corporate Logo



Who they are

Acer is based in New Taipei City, Taiwan. It was founded in 1976 as Multitech. In 1987, Stan Shih renamed it Acer. Revenue in 2025 was about \$9.6 billion. That's a lot of laptops and desktops. Acer doesn't make the CPU or RAM – they buy those from Intel, AMD, Samsung, etc. What they do is design the motherboard, the cooling system, the chassis, and then assemble everything. They also write the BIOS.

Acer owns factories in China and Taiwan, and also works with partners like Quanta. They're the quiet giant behind millions of computers that just work.

What they make

Laptops

- **Aspire** – the everyday home and office laptop. Affordable, reliable, no drama.
- **Swift** – thin and light, some with OLED screens. Great for students.
- **Predator** – gaming laptops with fast screens (240 Hz) and powerful GPUs (RTX 4080). They even use liquid metal for cooling.
- **ConceptD** – for creators who need accurate colors and ISV certifications for CAD software.

Figure 4: Acer Swift Lite 14 – Ultra-Lightweight OLED Laptop with Intel Core Ultra 5



Desktops and monitors

- **Aspire TC** – small tower for home or office.
- **Predator Orion 9000** – a massive gaming desktop that can hold two RTX 4090s.
- **Veriton** – small business desktops.
- **Monitors** – from budget office screens to high-end gaming displays with G-Sync.

How they build a computer

Acer starts by choosing components. CPUs from Intel or AMD, GPUs from NVIDIA (or AMD for some desktops), RAM from Samsung or SK Hynix, SSDs from Kingston or Western Digital, displays from BOE or LG. Then they design the motherboard to fit everything, add a cooling solution (fans, heat pipes, sometimes liquid metal), and choose a chassis (plastic or metal). They write the BIOS (using Insyde or AMI software) and add their own features like fan curves and Secure Boot. Finally, they assemble it, install Windows (or ChromeOS), and test it. It's a lot of work, but they make it look easy.

Operating system support

Most Acer laptops come with Windows 11 Home. For schools, they offer Windows 11 SE and Chromebooks. A very small number come with Ubuntu Linux. Acer doesn't provide official Linux drivers, but they make sure the hardware follows ACPI standards, so Linux usually works fine. In fact, many people buy Acer laptops specifically to learn Linux because they are affordable and well-supported by the community. That's pretty cool.

Why Acer matters

You might not think of Acer as exciting. But they sell millions of computers to people who just need something that works. Their Predator line shows they can also build

high-end gaming machines. And their ConceptD series proves they care about professionals. They are the reliable friend who never says no.

Task 3: G.Skill – The memory speed specialists

Figure 5: G.Skill Corporate Logo



Who they are

G.Skill is in Taipei, Taiwan. Founded in 2003. The CEO is Albert Lin. They only have about 200 employees, but they are famous among PC enthusiasts. Here's the key: G.Skill does **not** make memory chips. They buy chips from Samsung, SK Hynix, and Micron. Then they test them, select the fastest ones, put them on their own circuit boards, add heat spreaders (and sometimes RGB lights), and sell them as high-performance memory modules. Think of them as the quality control fanatics of the RAM world.

What they make

The table below shows their main product lines.

Series	Type	Speed (MT/s)	CAS Latency	Best for
Trident Z5 RGB	DDR5	6000–8400	30–40	Gaming with RGB
Trident Z5 Royal	DDR5	7200–8000	34–38	Showpiece builds
Flare X5	DDR5	5600–6000	30–36	AMD Ryzen (EXPO)
Ripjaws V	DDR4	2133–4800	14–19	Budget or older PCs

Figure 6: G.Skill Trident Z5 Royal Neo DDR5 Memory Module (Silver Crystalline Heat Spreader)



How they pick the best chips

Every DRAM chip that comes from Samsung or Micron is not the same. Some can run faster than others. G.Skill's engineers test thousands of chips. They increase the frequency step by step, measure stability, and also test at high temperatures (85°C for 24 hours). The very best chips – about the top 1% – go into the Trident Z Royal line. The rest go into lower-tier products. It's like hand-picking the best apples for a premium pie.

What is XMP?

XMP stands for Extreme Memory Profile. It's a small file stored on the memory stick. When you go into BIOS and enable XMP, the motherboard reads that file and automatically runs the RAM at a higher speed (and sometimes higher voltage) than the standard JEDEC spec. G.Skill programs two profiles: one for the highest stable speed, and another for slightly lower speed with tighter timings. XMP is overclocking, so it's not guaranteed to work on every CPU or motherboard – but G.Skill tests heavily to make it work for most. It's like a cheat code for your RAM.

Does the OS care?

No. The OS just sees standard DDR4 or DDR5 memory. But the extra speed makes a real difference. In video editing, going from 4800 MT/s to 7200 MT/s can cut render times by 8–12%. In some games, you get 5–10% more frames per second. The OS memory manager doesn't know why it's faster – it just enjoys the extra bandwidth. Kind of like driving on an empty highway vs. a crowded city street.

Why people love G.Skill

They are obsessive about speed. They hold world records in memory overclocking (over 11000 MT/s with liquid nitrogen). For a normal user, it's overkill. But for a PC enthusiast, buying G.Skill Trident Z Royal is like buying a hand-crafted watch – you appreciate the care that went into it.

Task 4: ADATA – The storage brand that just works

Figure 7: ADATA Corporate Logo



Who they are

ADATA is also from New Taipei City, Taiwan. Founded in 2001. CEO Simon Chen. Revenue about \$3.2 billion. They have two brands: ADATA for mainstream, and XPG for gaming/enthusiast. Like G.Skill, ADATA does not make the memory chips (NAND flash). They buy from Samsung, Micron, Kioxia, and SK Hynix. Then they design controllers, write firmware, assemble SSDs, and sell them. They're the invisible workhorse behind many fast computers.

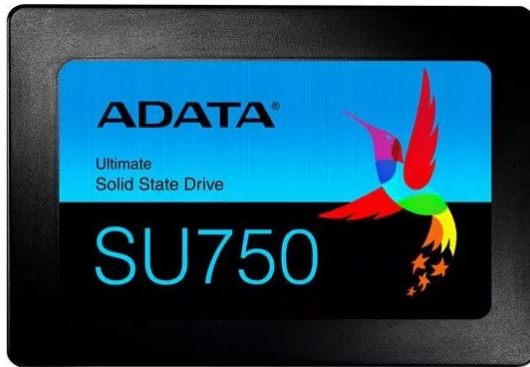
What they make

SSDs (internal)

Here are some of their most popular models.

Model	Interface	Max Read	Max Write	What's special
XPG GAMMIX S70 Blade	PCIe 4.0	7400 MB/s	6800 MB/s	In-house controller, includes heat sink
XPG GAMMIX S50 Core	PCIe 4.0	3900 MB/s	3200 MB/s	Budget PCIe 4.0
Legend 960	PCIe 5.0	10000 MB/s	9500 MB/s	Future-proof, very fast
SU800	SATA	560 MB/s	520 MB/s	Great for old laptop upgrades

Figure 8: ADATA Ultimate SU750 Solid State Drive – Reliable SATA Storage



External storage

- **SE920** – USB4 external SSD, 4000 MB/s, has a built-in fan.
- **SE800** – USB 3.2, 1000 MB/s, IP68 waterproof.
- **DashDrive** – USB flash drives from 32 GB to 1 TB.

How an SSD works (simply)

An SSD has two main parts: the NAND flash chips (where data is stored) and the controller (a tiny computer that manages the storage). ADATA designs the controller or buys one from Silicon Motion or Realtek. The controller does wear leveling (making sure no single memory cell wears out too fast), garbage collection (cleaning up deleted files in the background), and TRIM (responding to the OS when you delete a file). Good firmware makes the drive last longer and stay fast.

Operating system support

You don't need special drivers for ADATA SSDs. They follow the NVMe and SATA standards, so the OS uses built-in drivers. TRIM support is automatic on Windows 10/11 and modern Linux. On macOS, you have to enable it manually with trimforce enable. Once installed, an ADATA SSD can make a five-year-old laptop feel new again. Boot times drop from 90 seconds to 15 seconds. That's the kind of upgrade people actually notice – and smile about.

Reliability

The XPG S70 Blade 1 TB is rated for 740 TBW (terabytes written) – that means you could write 740,000 GB of data before it might fail. It has a 5-year warranty. The cheaper Legend 800 has 400 TBW and a 3-year warranty. For most people, these numbers are far beyond what they will ever use. You can stop worrying.

How ADATA compares

- **Samsung** – faster and more trusted, but more expensive.
- **Western Digital** – similar performance, but ADATA often cheaper.
- **Kingston** – ADATA has wider retail presence.
- **Corsair** – ADATA is more budget-friendly.

If you want a good SSD without spending too much, ADATA is a solid choice. No shame in saving money for the same real-world speed.

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Putting it all together: how they work with your OS

Imagine you have a laptop with a Qualcomm CPU, G.Skill RAM, an ADATA SSD, and it's assembled by Acer. You click a file. Here's what happens under the hood:

1. The OS sends a read command to the ADATA SSD.
2. The SSD controller finds the data and sends it to the G.Skill RAM (via DMA).
3. The Qualcomm CPU reads the data from RAM through its memory controller.
4. The OS scheduler decides which CPU core (performance or efficiency) should process the data.
5. The result is displayed on the screen – thanks to Acer's motherboard, BIOS, and cooling, all of which keep the system stable.

None of these companies builds the whole machine alone. But together, they create the experience we call computing. It's like a band – each member plays a different instrument, but the music only works when they all show up.

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Final thoughts

We chose Qualcomm, Acer, G.Skill, and ADATA because they represent different layers of the hardware stack. Qualcomm thinks about wireless and efficiency. Acer thinks about putting parts together affordably. G.Skill obsesses over memory speed. ADATA makes storage fast and reliable. Each one has a human story – a group of engineers who wake up every day trying to make your digital life a little better.

That's the real point of this assignment. Not just memorizing names, but understanding that behind every spec sheet, there are people who care about what they build. And maybe, just maybe, that makes you care a little more too.

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